

Combining Decision Procedures

The expressions which arise in program manipulation often do not fall within any... naturally defined theories — they usually involve mixed terms containing functions and predicates from several theories.

— Greg Nelson and Derek C. Oppen

Simplification by Cooperating Decision Procedures, 1979

Chapters 7–9 consider decision procedures for theories that each formalize just one data type. Yet almost all formulae in Chapter 5 are formulae of union theories. For example, many assert facts in $T_{\mathbb{Z}} \cup T_{\mathbf{A}}$ about arrays of integers indexed by integers. Additionally, the decision procedure for the array property fragment of $T_{\mathbf{A}}^{\mathbb{Z}}$ that we discuss in Chapter 11 requires a procedure for the quantifier-free fragment of $T_{\mathbb{Z}} \cup T_{\mathbf{A}}$. Can we reuse the decision procedures of Chapters 7–9 to decide satisfiability of formulae in union theories, or must we invent a new procedure for each combination?

Fortunately, there is a general result for quantifier-free fragments of union theories that allows us to reuse the procedures. This chapter discusses the **Nelson-Oppen combination method** for constructing decision procedures for union theories from decision procedures for individual theories. Section 10.1 introduces the method and discusses its limitations. Then Section 10.2 presents a nondeterministic version, for which correctness is proved in Section 10.4; and Section 10.3 presents the more practical deterministic version.

In this chapter, decision procedures for individual theories apply just to quantifier-free fragments. We rely on Cooper’s method with all optimizations for considering quantifier-free $\Sigma_{\mathbb{Z}}$ -formulae. Procedures for the other theories already apply only to their quantifier-free fragments.

10.1 Combining Decision Procedures

Consider two theories T_1 and T_2 over signatures Σ_1 and Σ_2 , respectively. For the quantifier-free fragments of T_1 and T_2 , we have decision procedures P_1 and P_2 . How do we decide satisfiability in the quantifier-free fragment of $T_1 \cup T_2$?

Example 10.1. Consider the $(\Sigma_E \cup \Sigma_Z)$ -formula

$$F : 1 \leq x \wedge x \leq 2 \wedge f(x) \neq f(1) \wedge f(x) \neq f(2) .$$

Chapter 9 describes a decision procedure for T_E , while Chapter 7 presents a decision procedure for T_Z . We would like to combine these decision procedures to decide the $(T_E \cup T_Z)$ -satisfiability of F and other quantifier-free $(\Sigma_E \cup \Sigma_Z)$ -formulae. ■

The **Nelson-Open combination method** (N-O method) combines decision procedures for the quantifier-free fragments of several theories into one decision procedure for the quantifier-free fragment of the union theory. In our presentation of the N-O method, we usually discuss combining two theories and their decision procedures; however, the N-O method can combine an arbitrary number of theories and procedures. Additionally, we restrict ourselves to considering conjunctive formulae; however, the satisfiability of arbitrary (quantifier-free) formulae can be considered by converting to DNF and checking each disjunct.

Besides being restricted to quantifier-free formulae, the N-O method has two additional restrictions. First, the signatures Σ_1 and Σ_2 can only share equality $=$:

$$\Sigma_1 \cap \Sigma_2 = \{=\} .$$

Second, the theories T_1 and T_2 must be stably infinite.

A theory T with signature Σ is **stably infinite** if for every quantifier-free Σ -formula F , if F is T -satisfiable, then there exists some T -interpretation that satisfies F and has a domain of infinite cardinality. We illustrate this concept with two example theories.

Example 10.2. Consider the theory $T_{a,b}$ with signature

$$\Sigma_2 : \{a, b, =\} ,$$

where both a and b are constants, and axiom

$$1. \forall x. x = a \vee x = b \tag{two}$$

Because of axiom (two), every $T_{a,b}$ -interpretation $I : (D_I, \alpha_I)$ is such that the domain D_I has at most two elements: $|D_I| \leq 2$. Hence, $T_{a,b}$ is not stably infinite. ■

Example 10.3. We prove that T_E is stably infinite. Consider the T_E -satisfiable quantifier-free Σ_E -formula F with arbitrary satisfying T_E -interpretation $I : (D_I, \alpha_I)$ in which α_I maps $=$ to $=_I$. Let A be any infinite set disjoint from D_I . Then construct new interpretation $J : (D_J, \alpha_J)$:

- $D_J = D_I \cup A$

- $\alpha_J = \{= \mapsto =_J, \dots\}$, where for $v_1, v_2 \in D_J$,

$$v_1 =_J v_2 \stackrel{\text{def}}{=} \begin{cases} v_1 =_I v_2 & \text{if } v_1, v_2 \in D_I \\ \top & \text{if } v_1 \text{ is the same element as } v_2 \\ \perp & \text{otherwise} \end{cases}$$

J is a T_E -interpretation satisfying F with infinite domain. Hence, T_E is stably infinite. ■

The other theories discussed in this book are also stably infinite.

Example 10.4. Consider the quantifier-free conjunctive $(\Sigma_E \cup \Sigma_Z)$ -formula

$$F : 1 \leq x \wedge x \leq 2 \wedge f(x) \neq f(1) \wedge f(x) \neq f(2) .$$

The signatures of T_E and T_Z only share $=$. Also, both theories are stably infinite. Hence, the N-O combination of the decision procedures for T_E and T_Z decides the $(T_E \cup T_Z)$ -satisfiability of F .

Intuitively, F is $(T_E \cup T_Z)$ -unsatisfiable. For the first two literals imply $x = 1 \vee x = 2$ so that $f(x) = f(1) \vee f(x) = f(2)$. Yet the last two literals contradict this conclusion. ■

10.2 Nelson-Oppen Method: Nondeterministic Version

In this section, we discuss the nondeterministic version of the N-O method. While simple to present, it suffers from high complexity. Section 10.3 reformulates the method to be deterministic and efficient.

Consider a quantifier-free conjunctive $(\Sigma_1 \cup \Sigma_2)$ -formula F . The N-O method proceeds in two steps.

10.2.1 Phase 1: Variable Abstraction

The variable abstraction phase transforms a quantifier-free conjunctive formula F into two quantifier-free conjunctive formulae, a Σ_1 -formula F_1 and a Σ_2 -formula F_2 , such that F and $F_1 \wedge F_2$ are $(T_1 \cup T_2)$ -equisatisfiable. That is, F is $(T_1 \cup T_2)$ -satisfiable iff $F_1 \wedge F_2$ is $(T_1 \cup T_2)$ -satisfiable. F_1 and F_2 are linked via a set of shared variables.

For term t , let $\text{hd}(t)$ be the root symbol; *e.g.*, $\text{hd}(f(x)) = f$. Then for $i, j \in \{1, 2\}$ and $i \neq j$, repeat the following transformations as long as possible:

1. if function $f \in \Sigma_i$ and $\text{hd}(t) \in \Sigma_j$,

$$F[f(t_1, \dots, t, \dots, t_n)] \implies F[f(t_1, \dots, w, \dots, t_n)] \wedge w = t$$

2. if predicate $p \in \Sigma_i$ and $\text{hd}(t) \in \Sigma_j$,

$$F[p(t_1, \dots, t, \dots, t_n)] \implies F[p(t_1, \dots, w, \dots, t_n)] \wedge w = t$$

3. if $\text{hd}(s) \in \Sigma_i$ and $\text{hd}(t) \in \Sigma_j$,

$$F[s = t] \implies F[w = t] \wedge w = s$$

w is a fresh variable in each application of a transformation. Transformation 3 also applies to $s \neq t$ literals: replace $F[s \neq t]$ with $F[w \neq t] \wedge w = s$.

After applying the transformations, each literal of the resulting formula falls entirely within the signature of one of the two theories (or possibly within each if it is just an equality $x = y$ or a disequality $x \neq y$ between variables: such literals are in every signature since they do not have symbols other than $=$). Divide the literals into two sets, one for each theory. These sets are not disjoint when there is a literal that is an equality or disequality between variables. Then return the conjunction of each set.

Example 10.5. Consider $(\Sigma_E \cup \Sigma_Z)$ -formula

$$F : 1 \leq x \wedge x \leq 2 \wedge f(x) \neq f(1) \wedge f(x) \neq f(2) .$$

Since $f \in \Sigma_=_$ and $1 \in \Sigma_Z$, replace $f(1)$ by $f(w_1)$ and add $w_1 = 1$ by transformation 1. Similarly, replace $f(2)$ by $f(w_2)$ and add $w_2 = 2$.

Now, the literals

$$1 \leq x, x \leq 2, w_1 = 1, \text{ and } w_2 = 2$$

are T_Z -literals, while the literals

$$f(x) \neq f(w_1) \text{ and } f(x) \neq f(w_2)$$

are T_E -literals. Hence, construct the Σ_Z -formula

$$F_Z : 1 \leq x \wedge x \leq 2 \wedge w_1 = 1 \wedge w_2 = 2$$

and the Σ_E -formula

$$F_E : f(x) \neq f(w_1) \wedge f(x) \neq f(w_2) .$$

F_Z and F_E share the variables x, w_1 , and w_2 . $F_Z \wedge F_E$ is $(T_E \cup T_Z)$ -equisatisfiable to F . ■

Example 10.6. Consider the $(\Sigma_E \cup \Sigma_Z)$ -formula

$$F : f(x) = x + y \wedge x \leq y + z \wedge x + z \leq y \wedge y = 1 \wedge f(x) \neq f(2) .$$

Intuitively, F is $(T_E \cup T_Z)$ -satisfiable: consider an interpretation in which $x = 0, y = 1, z = 1, f(0) = 1$, and $f(2) = 2$.

In the first literal, $\text{hd}(f(x)) = f \in \Sigma_E$ and $\text{hd}(x + y) = + \in \Sigma_Z$; thus, by transformation 3, replace the literal with

$$w_1 = x + y \wedge w_1 = f(x) .$$

In the last literal, $f \in \Sigma_E$ but $2 \in \Sigma_Z$, so by transformation 1, replace it with

$$f(x) \neq f(w_2) \wedge w_2 = 2 .$$

Now, separating the literals results in two formulae:

$$F_Z : w_1 = x + y \wedge x \leq y + z \wedge x + z \leq y \wedge y = 1 \wedge w_2 = 2$$

is a Σ_Z -formula, and

$$F_E : w_1 = f(x) \wedge f(x) \neq f(w_2)$$

is a Σ_E -formula. The conjunction $F_Z \wedge F_E$ is $(T_E \cup T_Z)$ -equisatisfiable to F . ■

10.2.2 Phase 2: Guess and Check

Phase 1 separates $(\Sigma_1 \cup \Sigma_2)$ -formula F into two formulae, Σ_1 -formula F_1 , and Σ_2 -formula F_2 . F_1 and F_2 are linked by a set of shared variables. Let

$$V = \text{shared}(F_1, F_2) = \text{free}(F_1) \cap \text{free}(F_2)$$

be the shared variables of F_1 and F_2 . Let E be an equivalence relation over V . The **arrangement** $\alpha(V, E)$ of V induced by E is the formula

$$\alpha(V, E) : \bigwedge_{u, v \in V. uEv} u = v \wedge \bigwedge_{u, v \in V. \neg(uEv)} u \neq v ,$$

which asserts that variables related by E are equal and that variables unrelated by E are not equal. The formula F is $(T_1 \cup T_2)$ -satisfiable iff there exists an equivalence relation E of V such that

- $F_1 \wedge \alpha(V, E)$ is T_1 -satisfiable, and
- $F_2 \wedge \alpha(V, E)$ is T_2 -satisfiable.

Otherwise, F is $(T_1 \cup T_2)$ -unsatisfiable.

Example 10.7. Consider $(\Sigma_E \cup \Sigma_Z)$ -formula

$$F : 1 \leq x \wedge x \leq 2 \wedge f(x) \neq f(1) \wedge f(x) \neq f(2) .$$

Phase 1 separates this formula into the Σ_Z -formula

$$F_Z : 1 \leq x \wedge x \leq 2 \wedge w_1 = 1 \wedge w_2 = 2$$

and the Σ_E -formula

$$F_E : f(x) \neq f(w_1) \wedge f(x) \neq f(w_2) ,$$

with

$$V = \text{shared}(F_Z, F_E) = \{x, w_1, w_2\} .$$

There are 5 equivalence relations to consider, which we list by stating the partitions:

1. $\{\{x, w_1, w_2\}\}$, *i.e.*, $x = w_1 = w_2$: $F_E \wedge \alpha(V, E)$ is T_E -unsatisfiable because it cannot be the case that both $x = w_1$ and $f(x) \neq f(w_1)$.
2. $\{\{x, w_1\}, \{w_2\}\}$, *i.e.*, $x = w_1$, $x \neq w_2$: $F_E \wedge \alpha(V, E)$ is T_E -unsatisfiable because it cannot be the case that both $x = w_1$ and $f(x) \neq f(w_1)$.
3. $\{\{x, w_2\}, \{w_1\}\}$, *i.e.*, $x = w_2$, $x \neq w_1$: $F_E \wedge \alpha(V, E)$ is T_E -unsatisfiable because it cannot be the case that both $x = w_2$ and $f(x) \neq f(w_2)$.
4. $\{\{x\}, \{w_1, w_2\}\}$, *i.e.*, $x \neq w_1$, $w_1 = w_2$: $F_Z \wedge \alpha(V, E)$ is T_Z -unsatisfiable because it cannot be the case that both $w_1 = w_2$ and $w_1 = 1 \wedge w_2 = 2$.
5. $\{\{x\}, \{w_1\}, \{w_2\}\}$, *i.e.*, $x \neq w_1$, $x \neq w_2$, $w_1 \neq w_2$: $F_Z \wedge \alpha(V, E)$ is T_Z -unsatisfiable because it cannot be the case that both $x \neq w_1 \wedge x \neq w_2$ and $x = w_1 = 1 \vee x = w_2 = 2$ (since $1 \leq x \leq 2$ implies that $x = 1 \vee x = 2$ in T_Z).

Hence, F is $(T_E \cup T_Z)$ -unsatisfiable. ■

Example 10.8. Consider the $(\Sigma_{\text{cons}} \cup \Sigma_Z)$ -formula

$$F : \text{car}(x) + \text{car}(y) = z \wedge \text{cons}(x, z) \neq \text{cons}(y, z) .$$

After two applications of transformation 1, Phase 1 separates F into the Σ_{cons} -formula

$$F_{\text{cons}} : w_1 = \text{car}(x) \wedge w_2 = \text{car}(y) \wedge \text{cons}(x, z) \neq \text{cons}(y, z)$$

and the Σ_Z -formula

$$F_Z : w_1 + w_2 = z ,$$

with

$$V = \text{shared}(F_{\text{cons}}, F_Z) = \{z, w_1, w_2\} .$$

Consider the equivalence relation E given by the partition

$$\{\{z\}, \{w_1\}, \{w_2\}\} .$$

The arrangement

$$\alpha(V, E) : z \neq w_1 \wedge z \neq w_2 \wedge w_1 \neq w_2$$

satisfies both F_{cons} and F_Z : $F_{\text{cons}} \wedge \alpha(V, E)$ is T_{cons} -satisfiable, and $F_Z \wedge \alpha(V, E)$ is T_Z -satisfiable. Hence, F is $(T_{\text{cons}} \cup T_Z)$ -satisfiable. ■

10.2.3 Practical Efficiency

Phase 2 is formulated as “guess and check”: first, guess an equivalence relation E , then check the induced arrangement. Unfortunately, the number of equivalence relations increases significantly with the number of shared variables. The

number of equivalence relations is given by the sequence of **Bell numbers**, which grows super-exponentially. For example, just 12 shared variables induce over four million equivalence relations. Hence, the guess-and-check method is impractical.

However, there is no need to guess the entire equivalence relation at once; instead, construct it incrementally, as the following example illustrates:

Example 10.9. In Example 10.6, Phase 1 separates the $(\Sigma_E \cup \Sigma_Z)$ -formula

$$F : f(x) = x + y \wedge x \leq y + z \wedge x + z \leq y \wedge y = 1 \wedge f(x) \neq f(2)$$

into Σ_Z -formula

$$F_Z : w_1 = x + y \wedge x \leq y + z \wedge x + z \leq y \wedge y = 1 \wedge w_2 = 2$$

and Σ_E -formula

$$F_E : w_1 = f(x) \wedge f(x) \neq f(w_2)$$

Then

$$V = \text{shared}(F_Z, F_E) = \{x, w_1, w_2\} .$$

We attempt to construct an arrangement.

1. Suppose $x = w_1$. But then $w_1 = x + y$ of F_Z implies that $y = 0$, yet F_Z asserts that $y = 1$. Hence, $x \neq w_1$.
2. $F_Z \wedge x \neq w_1$ and $F_E \wedge x \neq w_1$ are T_Z - and T_E -satisfiable, respectively.
3. Suppose $x = w_2$. But $f(x) \neq f(w_2)$ of F_E contradicts this supposition. Hence, $x \neq w_2$.
4. $F_Z \wedge x \neq w_1 \wedge x \neq w_2$ and $F_E \wedge x \neq w_1 \wedge x \neq w_2$ are T_Z - and T_E -satisfiable, respectively.
5. Suppose $w_1 = w_2$. No contradiction exists.

We discovered the arrangement

$$x \neq w_1 \wedge x \neq w_2 \wedge w_1 = w_2 ,$$

so F is $(T_E \cup T_Z)$ -satisfiable. ■

Readers interested in implementing a simple Nelson-Oppen-based decision procedure could consider this incremental-construction “optimization” of the nondeterministic method. However, in practice, implementations are based on the deterministic method described in the next section.

10.3 Nelson-Oppen Method: Deterministic Version

Phase 1 of the deterministic version is the same as in the nondeterministic version.

Phase 2 of the nondeterministic method (both the guess-and-check method and the optimized incremental construction) proposes a set of equalities and disequalities and then lets each decision procedure P_i check the set with the corresponding formula F_i . In contrast, Phase 2 of the deterministic version asks the decision procedures P_1 and P_2 to propagate information in the form of new equalities.

A **convex** theory is particularly well-suited for propagating equalities. Section 10.3.1 discusses convex theories. Then Section 10.3.2 presents the deterministic Nelson-Oppen method.

10.3.1 Convex Theories

If a conjunctive formula in a convex theory implies a disjunction of equalities between variables, then it actually implies a single equality. Formally, consider a quantifier-free conjunctive Σ -formula F and a disjunction

$$G : \bigvee_{i=1}^n u_i = v_i , \quad (10.1)$$

for variables u_i and v_i . Theory T is **convex** if for every such F and G , if

$$F \Rightarrow \bigvee_{i=1}^n u_i = v_i$$

then

$$F \Rightarrow u_i = v_i \quad \text{for some } i \in \{1, \dots, n\} .$$

If F implies G , then F actually implies one of the disjuncts of G .

Intuitively, F cannot be “covered” by any disjunction of equalities — no matter how many — if no single equality covers F (F is covered by a formula if F implies it). This intuition is especially apparent for vector spaces (Section 8.2): a plane cannot be covered by a finite disjunction of lines; it cannot even be covered by a finite disjunction of other planes unless at least one of the planes is the plane itself.

Example 10.10. The theory of integers $T_{\mathbb{Z}}$ is not convex. For consider the quantifier-free conjunctive $\Sigma_{\mathbb{Z}}$ -formula

$$F : 1 \leq z \wedge z \leq 2 \wedge u = 1 \wedge v = 2 .$$

Then

$$F \Rightarrow z = u \vee z = v ,$$

but neither

$$F \Rightarrow z = u \quad \text{nor} \quad F \Rightarrow z = v .$$

■

Example 10.11. The theory of arrays T_A is not convex. For consider the quantifier-free conjunctive Σ_A -formula

$$F : a\langle i \triangleleft v \rangle[j] = v .$$

Then

$$F \Rightarrow i = j \vee a[j] = v ,$$

but neither

$$F \Rightarrow i = j \quad \text{nor} \quad F \Rightarrow a[j] = v .$$

■

Example 10.12. ★ The theory of rationals $T_{\mathbb{Q}}$ is convex, as it is convex in a geometric sense (see Chapter 8).

Each equality $u_i = v_i$ of the disjunction G of (10.1) is geometrically convex, but G itself is not. Consider, for example,

$$H : x = y \vee x = z .$$

Let S_H be the set of points satisfying H . The point $(x, y, z) = (0, 0, 1)$ is included in S_H , as is the point $(1, 0, 1)$. However, the average of the two points, $(\frac{1}{2}, 0, 1)$ (choosing $\lambda = \frac{1}{2}$), is not in S_H . Indeed, choose any two points

$$(u, u, v_1) \quad \text{and} \quad (w, v_2, w)$$

from $S_{x=y}$ and $S_{x=z}$, respectively, such that neither is in their intersection $S_{x=y=z}$ (i.e., $v_1 \neq u$ and $v_2 \neq w$). Then for any $\lambda \in (0, 1)$, the point

$$(\lambda u + (1 - \lambda)w, \lambda u + (1 - \lambda)v_2, \lambda v_1 + (1 - \lambda)w)$$

is neither in $S_{x=y}$ nor in $S_{x=z}$.

Suppose, then, that $F \Rightarrow G : \bigvee_{i=1}^n u_i = v_i$, but for no $i \in \{1, \dots, n\}$ does $F \Rightarrow u_i = v_i$. Then it must be the case that there are two points s_1 and s_2 of S_F in separate subsets $S_{u_i=v_i}$, $i \neq j$, of S_G . By the argument above, the points on the line segment between s_1 and s_2 are not in S_G and thus not in S_F . Then F is not geometrically convex, a contradiction.

Thus, $T_{\mathbb{Q}}$ is convex. ■

Exercise 10.5 asks the reader to prove that the theories T_E and T_{cons} are also convex.

10.3.2 Phase 2: Equality Propagation

Recall that the nondeterministic version guesses an equivalence relation E over the shared variables V and checks that both $F_1 \wedge \alpha(V, E)$ is T_1 -satisfiable and $F_2 \wedge \alpha(V, E)$ is T_2 -satisfiable. If it finds a satisfying equivalence relation E , it declares that F is $(T_1 \cup T_2)$ -satisfiable. This method suffers from the enormous number of equivalence relations that are possible even over small sets of shared variables. In the deterministic version, a central manager asks the decision procedures P_1 and P_2 to report any new implied equalities between shared variables. It then adds this new information to the already discovered equalities and propagates it to the other decision procedure. This method is efficient.

In the context of already discovered equalities $\mathcal{E}$, a decision procedure P_i for a convex theory T_i discovers a new equality $u = v$, for shared variables u and v , when

$$F_i \wedge \mathcal{E} \Rightarrow u = v .$$

The central manager then propagates this new equality to the other decision procedure.

If T_j is not convex, P_j discovers a new disjunction of equalities S when

$$F_j \wedge \mathcal{E} \Rightarrow \bigvee_{u_i=v_i \in S} (u_i = v_i) ,$$

for shared variables u_i and v_i . In this case, the central manager must split the disjunction and search along multiple branches. Each branch assumes one of the disjuncts. The search along a branch ends either when a full arrangement is discovered (so the original formula is $(T_1 \cup T_2)$ -satisfiable; see below) or when all sub-branches end in contradiction (T_i -unsatisfiability for some i). In the latter case, the central manager tries another branch. If no branches remain to try, then the central manager declares the original formula to be $(T_1 \cup T_2)$ -unsatisfiable.

If at some point, neither P_1 nor P_2 finds a new equality (or a disjunction of equalities in the non-convex case), then the central manager concludes that the given formula is $(T_1 \cup T_2)$ -satisfiable. For if $\mathcal{E}$ is the set of all learned equalities, S is the set of all possible remaining equalities, and

$$F_1 \wedge \mathcal{E} \not\Rightarrow \bigvee_{u_i=v_i \in S} (u_i = v_i) \quad \text{and} \quad F_2 \wedge \mathcal{E} \not\Rightarrow \bigvee_{u_i=v_i \in S} (u_i = v_i) ,$$

(which must hold when no new disjunctions of equalities are discovered), then

$$F_1 \wedge \mathcal{E} \wedge \bigwedge_{u_i=v_i \in S} (u_i \neq v_i) \quad \text{and} \quad F_2 \wedge \mathcal{E} \wedge \bigwedge_{u_i=v_i \in S} (u_i \neq v_i)$$

are T_1 -satisfiable and T_2 -satisfiable, respectively. Hence, the discovered arrangement is

$$\alpha(V, E) = \mathcal{E} \wedge \bigwedge_{u_i=v_i \in S} (u_i \neq v_i) ,$$

and F is $(T_1 \cup T_2)$ -satisfiable.

Example 10.13. Consider the $(\Sigma_E \cup \Sigma_Q)$ -formula

$$F : f(f(x) - f(y)) \neq f(z) \wedge x \leq y \wedge y + z \leq x \wedge 0 \leq z .$$

F is $(T_E \cup T_Q)$ -unsatisfiable: the final three literals imply that $z = 0$ and $x = y$, so that $f(x) = f(y)$. But then from the first literal, $f(0) \neq f(0)$ since both $f(x) - f(y)$ and z equal 0.

Phase 1 separates F into two formulae. According to transformation 1, it replaces $f(x)$ by u , $f(y)$ by v , and $u - v$ by w , resulting in Σ_E -formula

$$F_E : f(w) \neq f(z) \wedge u = f(x) \wedge v = f(y)$$

and Σ_Q -formula

$$F_Q : x \leq y \wedge y + z \leq x \wedge 0 \leq z \wedge w = u - v ,$$

with

$$V = \text{shared}(F_E, F_Q) = \{x, y, z, u, v, w\} .$$

Recall that T_E and T_Q are convex theories. The decision procedure P_Q for T_Q discovers

$$F_Q \Rightarrow x = y$$

from $x \leq y \wedge y + z \leq x \wedge 0 \leq z$, so

$$\mathcal{E}_1 : x = y .$$

Then P_E discovers the new congruence $f(x) = f(y)$ from $x = y$, so that

$$F_E \wedge \mathcal{E}_1 \Rightarrow u = v ,$$

yielding

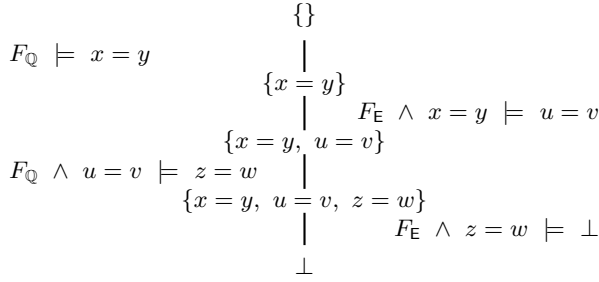
$$\mathcal{E}_2 : x = y \wedge u = v .$$

But then

$$F_Q \wedge \mathcal{E}_2 \Rightarrow z = w$$

since $w = u - v = 0$, according to $u = v$, and $z = 0$. Propagating this equality back to P_E via

$$\mathcal{E}_3 : x = y \wedge u = v \wedge z = w$$

**Fig. 10.1.** Summary of Example 10.13

reveals the contradiction

$$F_{\mathbb{E}} \wedge \mathcal{E}_3 \Rightarrow \perp ;$$

in particular, $z = w$ contradicts $f(w) \neq f(z)$. Therefore, F is $(T_{\mathbb{E}} \cup T_{\mathbb{Q}})$ -unsatisfiable.

Since both $T_{\mathbb{E}}$ and $T_{\mathbb{Q}}$ are convex, no case splitting was required.

Figure 10.1 summarizes this argument. The left and right halves list deductions made in $T_{\mathbb{Q}}$ and $T_{\mathbb{E}}$, respectively. The sets in the middle are the deduced sets of shared equalities. The deductions terminate with $\perp$, indicating that F is $(T_{\mathbb{E}} \cup T_{\mathbb{Q}})$ -unsatisfiable. ■

Example 10.14. Consider the $(\Sigma_{\mathbb{E}} \cup \Sigma_{\mathbb{Z}})$ -formula

$$F : 1 \leq x \wedge x \leq 2 \wedge f(x) \neq f(1) \wedge f(x) \neq f(2) .$$

While $T_{\mathbb{E}}$ is convex, $T_{\mathbb{Z}}$ is not. Thus, we should expect some case splits.

According to transformation 1, Phase 1 replaces $f(1)$ by $f(w_1)$ and $f(2)$ by $f(w_2)$, resulting in the $\Sigma_{\mathbb{Z}}$ -formula

$$F_{\mathbb{Z}} : 1 \leq x \wedge x \leq 2 \wedge w_1 = 1 \wedge w_2 = 2$$

and the $\Sigma_{\mathbb{E}}$ -formula

$$F_{\mathbb{E}} : f(x) \neq f(w_1) \wedge f(x) \neq f(w_2) ,$$

with

$$V = \text{shared}(F_{\mathbb{Z}}, F_{\mathbb{E}}) = \{x, w_1, w_2\} .$$

Immediately, $P_{\mathbb{Z}}$ recognizes that

$$F_{\mathbb{Z}} \Rightarrow x = w_1 \vee x = w_2 ,$$

since $1 \leq x \leq 2$ implies that either $x = 1$ or $x = 2$. Hence, case split on these two disjuncts. For the first case, propagate

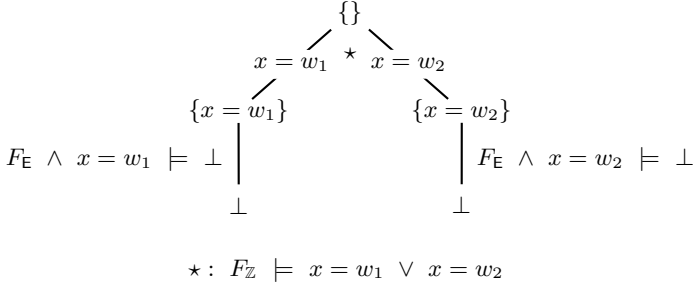


Fig. 10.2. Summary of Example 10.14

$$\mathcal{E}_1^a : x = w_1$$

to P_E , which discovers that

$$F_E \wedge \mathcal{E}_1^a \Rightarrow \perp ,$$

as $x = w_1$ contradicts $f(x) \neq f(w_1)$.

For the second case,

$$\mathcal{E}_1^b : x = w_2 .$$

Again, P_E discovers that

$$F_E \wedge \mathcal{E}_1^b \Rightarrow \perp ,$$

as $x = w_2$ contradicts $f(x) \neq f(w_2)$.

As all branches end in contradiction, F is $(T_E \cup T_{\mathbb{Z}})$ -unsatisfiable.

Figure 10.2 summarizes this argument. Unlike in Example 10.13 and Figure 10.1, the nonconvexity of $T_{\mathbb{Z}}$ causes the argument to branch along two possibilities. Each branch ends in a contradiction. ■

Example 10.15. Consider the $(\Sigma_E \cup \Sigma_{\mathbb{Z}})$ -formula

$$F : 1 \leq x \wedge x \leq 3 \wedge f(x) \neq f(1) \wedge f(x) \neq f(3) \wedge f(1) \neq f(2) .$$

Applying transformation 1 of Phase 1 three times produces the $\Sigma_{\mathbb{Z}}$ -formula

$$F_{\mathbb{Z}} : 1 \leq x \wedge x \leq 3 \wedge w_1 = 1 \wedge w_2 = 2 \wedge w_3 = 3$$

and the Σ_E -formula

$$F_E : f(x) \neq f(w_1) \wedge f(x) \neq f(w_3) \wedge f(w_1) \neq f(w_2) ,$$

with

$$V = \text{shared}(F_{\mathbb{Z}}, F_{\mathbb{E}}) = \{x, w_1, w_2, w_3\} .$$

From $1 \leq x \leq 3$, $P_{\mathbb{Z}}$ discovers that

$$F_{\mathbb{Z}} \Rightarrow x = w_1 \vee x = w_2 \vee x = w_3 .$$

Recall that $T_{\mathbb{Z}}$ is not convex. On case

$$\mathcal{E}_1^a : x = w_1 ,$$

$P_{\mathbb{E}}$ finds that

$$F_{\mathbb{E}} \wedge \mathcal{E}_1^a \Rightarrow \perp$$

because of $f(x) \neq f(w_1)$. On case

$$\mathcal{E}_1^b : x = w_2 ,$$

neither $P_{\mathbb{Z}}$ nor $P_{\mathbb{E}}$ discovers any contradiction or new equality. That is,

$$F_{\mathbb{Z}} \wedge \mathcal{E}_1^b \not\Rightarrow x = w_1 \vee x = w_3 \vee w_1 = w_2 \vee w_1 = w_3 \vee w_2 = w_3$$

and

$$F_{\mathbb{E}} \wedge \mathcal{E}_1^b \not\Rightarrow x = w_1 \vee x = w_3 \vee w_1 = w_2 \vee w_1 = w_3 \vee w_2 = w_3 ;$$

or, in other words,

$$F_{\mathbb{Z}} \wedge \mathcal{E}_1^b \wedge x \neq w_1 \wedge x \neq w_3 \wedge w_1 \neq w_2 \wedge w_1 \neq w_3 \wedge w_2 \neq w_3$$

is $T_{\mathbb{Z}}$ -satisfiable, and

$$F_{\mathbb{E}} \wedge \mathcal{E}_1^b \wedge x \neq w_1 \wedge x \neq w_3 \wedge w_1 \neq w_2 \wedge w_1 \neq w_3 \wedge w_2 \neq w_3$$

is $T_{\mathbb{E}}$ -satisfiable. Thus, F is $(T_{\mathbb{E}} \cup T_{\mathbb{Z}})$ -satisfiable.

Figure 10.3 summarizes this argument. The middle branch terminates with a satisfying arrangement. We did not actually explore the right branch. ■

10.3.3 Equality Propagation: Implementation

Equality propagation can be implemented somewhat efficiently without modifying the individual decision procedures. For convex theory T_j , test each possible equality $u_i = v_i$. Suppose that F_j is the Σ_j -formula constructed in Phase 1 and $\mathcal{E}$ is the conjunction of equalities discovered so far. Then check if any equality $u_i = v_i$ is implied:

$$F_j \wedge \mathcal{E} \Rightarrow u_i = v_i .$$

Any implied equality should be propagated to the other theories.

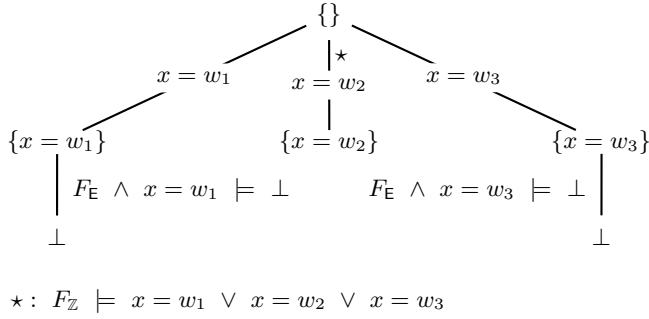


Fig. 10.3. Summary of Example 10.15

This procedure is not applicable to a non-convex theory T_k . A procedure for a non-convex theory must be able to find disjunctions of equalities that are implied by a Σ_k -formula F_k . Moreover, the disjunctions should be as small as possible since the Nelson-Oppen method must branch on each disjunct. A disjunction is **minimal** if it is implied by F_k and if each smaller disjunction is not implied by F_k .

A simple procedure to find a minimal disjunction is based on the observation that any disjunction that contains a minimal disjunction — which is implied by F_k by definition — is also implied by F_k . Therefore, we can strip off extra disjuncts one-by-one. First, consider the disjunction of all equalities at once. If it is not implied, then no subset is implied either, so we are done. Otherwise, drop each equality in turn: if the remaining disjunction is still implied by F_k , continue with this smaller disjunction; otherwise, restore the equality and continue. When all equalities have been considered, the resulting disjunction is minimal. This procedure requires checking T_k -satisfiability $O(|V|^2)$ times, where V is the set of shared variables. Exercise 10.4 asks the reader to describe a procedure based on binary search that requires asymptotically fewer satisfiability checks when the final disjunction is small relative to the disjunction of all equalities.

10.4 ★Correctness of the Nelson-Oppen Method

In this section, we prove the correctness of the Nelson-Oppen combination method. We reason at the level of arrangements, which is more suited to the nondeterministic version of the method. However, Section 10.3 shows how to construct an arrangement in the deterministic version, as well, so the following proof can be extended to the deterministic version. We also focus on the second phase of the nondeterministic procedure, which chooses an arrangement if one exists. We thus assume that the variable abstraction phase is correct: it

produces formulae F_1 and F_2 such that $F_1 \wedge F_2$ is $(T_1 \cup T_2)$ -equivalent to the given $(\Sigma_1 \cup \Sigma_2)$ -formula F .

A theory **has equality** (or is a theory **with equality**) if its signature includes the binary predicate $=$ and its axioms imply reflexivity, symmetry, and transitivity of equality. The **pure equality** fragment of a theory with equality is composed of formulae that are possibly quantified Boolean combinations of equalities between variables.

Theorem 10.16 (Sound & Complete). *Consider stably infinite theories T_1 and T_2 such that $\Sigma_1 \cap \Sigma_2 = \{=\}$. For conjunctive quantifier-free Σ_1 -formula F_1 and conjunctive quantifier-free Σ_2 -formula F_2 , $F_1 \wedge F_2$ is $(T_1 \cup T_2)$ -satisfiable iff there exists an arrangement $K = \alpha(\text{shared}(F_1, F_2), E)$ such that $F_1 \wedge K$ is T_1 -satisfiable and $F_2 \wedge K$ is T_2 -satisfiable.*

Soundness is straightforward. Suppose that $F_1 \wedge F_2$ is $(T_1 \cup T_2)$ -satisfiable with satisfying $(T_1 \cup T_2)$ -interpretation I . Extract from I the equivalence relation E such that the arrangement $K = \alpha(\text{shared}(F_1, F_2), E)$ is satisfied by I . Then $F_1 \wedge K$ and $F_2 \wedge K$ are both satisfied by I , which can be viewed as both a T_1 -interpretation and a T_2 -interpretation, so that they are T_1 -satisfiable and T_2 -satisfiable, respectively.

Completeness is more complicated. Let $K = \alpha(\text{shared}(F_1, F_2), E)$ be an arrangement such that $F_1 \wedge K$ and $F_2 \wedge K$ are T_1 -satisfiable and T_2 -satisfiable, respectively. Suppose that $F_1 \wedge F_2$ is $(T_1 \cup T_2)$ -unsatisfiable. We derive a contradiction.

The outline of the proof is the following. Because $F_1 \wedge F_2$ is $(T_1 \cup T_2)$ -unsatisfiable, we know that F_1 implies $\neg F_2$ in $T_1 \cup T_2$. An adaptation of the **Craig Interpolation Lemma** (Theorem 2.38) tells us that there is a quantifier-free formula H such that F_1 implies H over all infinite T_1 -interpretations (T_1 -interpretations with infinite domains) and F_2 implies $\neg H$ over all infinite T_2 -interpretations: H interpolates between F_1 and F_2 . We then show that the arrangement K implies H , which means that F_2 implies $\neg K$ over all infinite T_2 -interpretations. In other words, no infinite T_2 -interpretation satisfies $F_2 \wedge K$. Yet if T_2 is stably infinite and $F_2 \wedge K$ is T_2 -satisfiable as assumed, then $F_2 \wedge K$ is satisfied by some infinite T_2 -interpretation, a contradiction.

We now present the details of the proof. First, because we are considering only stably infinite theories, we need only consider interpretations with infinite domains. For we can extend a T_1 - or T_2 -interpretation with a finite domain to a T_1 - or T_2 -interpretation with an infinite domain. Therefore, define $\Rightarrow^*$ as a weaker form of implication: $F \Rightarrow^* G$ iff G is true on every interpretation I that has an infinite domain and that satisfies F . Similarly, weaken $\Leftrightarrow$ to $\Leftrightarrow^*$. If $F \Rightarrow^* G$, we say that F **weakly implies** G ; if $F \Leftrightarrow^* G$, we say that F is **weakly equivalent** to G .

Recall from Section 2.7.4 the following theorem.

Theorem 10.17 (Compactness Theorem). *A countable set of first-order formulae S is simultaneously satisfiable iff the conjunction of every finite subset is satisfiable.*

Since $F_1 \wedge F_2$ is $(T_1 \cup T_2)$ -unsatisfiable, the Compactness Theorem tells us that there exist a conjunction S_1 of a finite subset of axioms of T_1 and a conjunction S_2 of a finite subset of axioms of T_2 such that $S_1 \wedge F_1 \wedge S_2 \wedge F_2$ is (first-order) unsatisfiable. Choose S_1 and S_2 to include the axioms that imply reflexivity, symmetry, and transitivity of equality. Then, rearranging, we have that

$$S_1 \wedge F_1 \Rightarrow \neg S_2 \vee \neg F_2 . \quad (10.2)$$

Recall from Section 2.7.4 the following theorem.

Theorem 10.18 (Craig Interpolation Lemma). *If $F_1 \Rightarrow F_2$, then there exists a formula H such that $F_1 \Rightarrow H$, $H \Rightarrow F_2$, and each free variable, function symbol, and predicate symbol of H appears in F_1 and F_2 .*

Hence, from implication (10.2), there exists an interpolant H' such that $\text{free}(H') = \text{shared}(F_1, F_2)$ and

$$S_1 \wedge F_1 \Rightarrow H' \quad \text{and} \quad S_2 \wedge H' \Rightarrow \neg F_2 .$$

The latter implication is derived by rearranging $H' \Rightarrow \neg S_2 \vee \neg F_2$. Because $=$ is the only predicate or function shared between $S_1 \wedge F_1$ and $S_2 \wedge F_2$, H' is of a special form: its atoms are equalities between variables of $\text{shared}(F_1, F_2)$. However, H' may have quantifiers. We prove next that in fact a “weak” quantifier-free interpolant H exists.

Lemma 10.19 (Weak Quantifier Elimination for Pure Equality). *Consider any stably infinite theory T with equality. For each pure equality formula F , there exists a quantifier-free pure equality formula F' such that F is weakly T -equivalent to F' .*

Proof. Consider pure equality formula $\exists x. G[x, \bar{y}]$, where G is quantifier-free with free variables x and $\bar{y}$. Define

$$G_0 : G\{x = x \mapsto \text{true}, x = y_1 \mapsto \text{false}, \dots, x = y_n \mapsto \text{false}\}$$

and, for $i \in \{1, \dots, n\}$,

$$G_i : G\{x \mapsto y_i\} .$$

We claim that $\exists x. G$ is weakly T -equivalent to

$$G' : G_0 \vee G_1 \vee \dots \vee G_n .$$

For G' asserts that x is either equal to some free variable y_i or not. Because we consider only interpretations with infinite domains, it is always possible for x not to equal any y_i .

By Section 7.1, we have a weak quantifier elimination procedure over the pure equality fragment of T . It is weak because equivalence is only guaranteed to hold on infinite interpretations. ■

Example 10.20. Consider the pure equality formula

$$F : x \neq y \wedge (\forall z. z = x \vee z = y) .$$

For eliminating z , consider the negation of the second conjunct,

$$G : \exists z. z \neq x \wedge z \neq y ,$$

for which we have

$$G_0 : \neg \perp \wedge \neg \perp \Leftrightarrow \top$$

and

$$G_x : x \neq x \wedge x \neq y \Leftrightarrow \perp \quad G_y : y \neq x \wedge y \neq y \Leftrightarrow \perp .$$

Then

$$G' : G_0 \vee G_x \vee G_y \Leftrightarrow \top .$$

Substituting into F , we have

$$x \neq y \wedge \neg(\top) \Leftrightarrow \perp .$$

Hence, over infinite interpretations satisfying the axioms of equality, F is equivalent to $\perp$.

However, in an interpretation with a two-element domain that satisfies the equality axioms, F is not equivalent to $\perp$, but rather to $x \neq y$. For if $x \neq y$ on such an interpretation, then every element is equal either to x or to y . ■

Continuing the main theorem, we claim that there exists a quantifier-free pure equality formula H over $\text{shared}(F_1, F_2)$ such that

$$S_1 \wedge F_1 \Rightarrow^* H \quad \text{and} \quad S_2 \wedge H \Rightarrow^* \neg F_2 .$$

For by Lemma 10.19, a quantifier-free pure equality formula H exists such that H is weakly equivalent to the Craig interpolant H' in any stably infinite theory with equality.

For the next step, recall from the beginning of the proof that $F_1 \wedge K$ is T_1 -satisfiable and $F_2 \wedge K$ is T_2 -satisfiable, where $K = \alpha(\text{shared}(F_1, F_2), E)$ is an arrangement. We thus know that

$$S_1 \wedge F_1 \wedge K \quad \text{and} \quad S_2 \wedge F_2 \wedge K$$

are (first-order) satisfiable. Moreover, as T_1 and T_2 are stably infinite, each of these formulae has an interpretation with an infinite domain.

Now, K is a conjunction of equalities and disequalities between pairs of variables of $\text{shared}(F_1, F_2)$. Moreover, by the definition of an arrangement, K is as strong as possible: no additional equality literals L over $\text{shared}(F_1, F_2)$

can be added to K without either K and $K \wedge L$ being equivalent in a theory with equality or $K \wedge L$ being unsatisfiable in a theory with equality. Based on this observation, construct the formula K' by conjoining additional equality literals: for each pair of variables $u, v \in \text{shared}(F_1, F_2)$, conjoin either $u = v$ or $u \neq v$, depending on which maintains the satisfiability of K' in a theory with equality. Now, since $S_1 \wedge F_1 \wedge K$ is satisfiable, then so is $S_1 \wedge F_1 \wedge K'$, indeed by the same interpretations.

We claim that the DNF representation of H must include K' or a (conjunctive) subformula of K' as a disjunct. Suppose not; then every disjunct of the DNF representation of H contradicts the satisfying interpretations of $S_1 \wedge F_1 \wedge K'$, of which at least one exists. Therefore, $K' \Rightarrow H$, and — because K and K' are equivalent in a theory with equality — $K \Rightarrow H$. In other words, the discovered arrangement K is a special case of the weak interpolant H .

To finish, we have

$$S_2 \wedge H \Rightarrow^* \neg F_2 ,$$

or, rearranging,

$$S_2 \wedge F_2 \Rightarrow^* \neg H .$$

From $K \Rightarrow H$, we have $\neg H \Rightarrow \neg K$, so

$$S_2 \wedge F_2 \Rightarrow^* \neg K .$$

But this weak implication contradicts that $S_2 \wedge F_2 \wedge K$ is satisfied by some infinite interpretation. Thus, $F_1 \wedge F_2$ is actually $(T_1 \cup T_2)$ -satisfiable, and the Nelson-Oppen method is correct.

10.5 ★Complexity

Assume that T_1 and T_2 are stably infinite theories such that $\Sigma_1 \cap \Sigma = \{=\}$. Also, they have decision procedures P_1 and P_2 for their respective conjunctive quantifier-free fragments.

Theorem 10.21 (Complexity: Convex Theories). *If convex theories T_1 and T_2 have PTIME decision procedures P_1 and P_2 , then the Nelson-Oppen combination based on equality propagation is a PTIME decision procedure for the conjunctive quantifier-free fragment of $T_1 \cup T_2$.*

Theorem 10.22 (Complexity: Non-Convex Theories). *If T_1 and T_2 have NPTIME decision procedures P_1 and P_2 , then the Nelson-Oppen combination based on equality propagation is an NPTIME decision procedure for the conjunctive quantifier-free fragment of $T_1 \cup T_2$.*

10.6 Summary

Combining decision procedures in a general and efficient manner is crucial for most applications. This chapter covers the Nelson-Oppen combination method, in particular:

- The *nondeterministic Nelson-Oppen method*. Three requirements: the theories only share $=$; the theories are stably infinite; and the considered formula is quantifier-free. Variable abstraction, separation into theory-specific formulae. Shared variables, equivalence relations over shared variables, arrangements.
- The *deterministic Nelson-Oppen method*. Convex theories. Equality propagation.
- *Correctness* of the Nelson-Oppen method, which follows from the Craig Interpolation Lemma of Chapter 2.
- *Complexity*. When the individual decision procedures are convex and run in polynomial time, the combination procedure runs in polynomial time.

The Nelson-Oppen combination method provides a general means of reasoning simultaneously about the theories studied in this book using the individual decision procedures. Being able to reason in union theories is crucial. For example, almost all of the verification conditions of Chapters 5 and 6 are expressed in multiple signatures.

Bibliographic Remarks

Nelson and Oppen describe the Nelson-Oppen combination method [65]. Their original proof of correctness was flawed; Oppen presents a corrected proof in [70], and Nelson presents a corrected proof in [64]. Oppen also proves in [70] the complexity results that we state. Tinelli and Harandi present an alternate proof of correctness in [92]. Our correctness proof derives from that of Nelson and Oppen. See [56] for another presentation of the method and its correctness.

Another general combination method that has received much attention is that of Shostak [84]. See the work of Ruess and Shankar [78] for a correct presentation of the method.

Exercises

10.1 (DP for combinations). For each of the following formulae, identify the combination of theories in which it lies. To avoid ambiguity, prefer $T_{\mathbb{Z}}$ to $T_{\mathbb{Q}}$. Then apply the N-O method using the appropriate decision procedures. Use either the nondeterministic or deterministic version. Provide a level of detail as in the examples of the chapter.

- (a) $1 \leq x \wedge x \leq 2 \wedge \text{cons}(1, y) \neq \text{cons}(x, y) \wedge \text{cons}(2, y) \neq \text{cons}(x, y)$
 (b) $a[i] \geq 1 \wedge a[i] + x \leq 2 \wedge x > 0 \wedge x = i \wedge a\langle x \triangleleft 2 \rangle[i] \neq 1$

10.2 (Deterministic N-O). Apply the deterministic N-O method to the following formulae. Prefer $T_{\mathbb{Q}}$ to $T_{\mathbb{Z}}$.

- (a) $1 \leq x \wedge x \leq 2 \wedge \text{cons}(1, y) \neq \text{cons}(x, y) \wedge \text{cons}(2, y) \neq \text{cons}(x, y)$
 (b) $x + y = z \wedge f(z) = z \wedge f(x + y) \neq z$
 (c) $g(x + y, z) = f(g(x, y)) \wedge x + z = y \wedge z \geq 0 \wedge x \geq y$
 $\wedge g(x, x) = z \wedge f(z) \neq g(2x, 0)$

10.3 (★Equality propagation in T_E). Section 10.3.3 explains general techniques for propagating equalities. However, some decision procedures are easily modified to propagate new equalities. Describe such a modification of the congruence closure algorithm of Chapter 9.

10.4 (★Equality propagation). Consider conjunctive Σ -formula F of non-convex theory T and the disjunction of equalities

$$G : \bigvee_{i=1}^n u_i = v_i$$

such that $F \Rightarrow G$. Describe a procedure based on binary search that discovers a minimal disjunction G' of the equalities of G that is implied by F . If the procedure returns a disjunction with m equalities, then it should have invoked the decision procedure for T at most $O(m \lg n)$ times. *Hint:* The solution is related to the solution of Exercise 8.1(e).

10.5 (★Convex theories). Prove that the following theories are convex:

- (a) T_E
 (b) T_{cons}

10.6 (★Complexity). Prove the complexity results about the N-O method.

- (a) Theorem 10.21.
 (b) Theorem 10.22.